

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)

QUESTIONS: 5

Total Marks:50

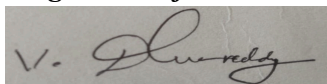
Annexure A SENARIO BASED QUESTIONS

<i>Criteria</i>	Understanding of Scenario (2)	Identification of Key Issues (2)	Applicatio n of Concepts (2.5)	Decision Making (2)	Clarity of Response (1.5)	<i>Total(10)</i>
<i>Weightage</i>	20%	20%	25%	20%	15%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I __v.dharanireddy(192311219)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A
Scenario based Questions

Q1. Data Preprocessing in Retail Analytics (L3, L4)

A **retail analytics company** is preparing customer purchase data for building a recommendation system. The dataset contains missing values, inconsistent categorical formats, and extreme values due to data collected from multiple stores and manual entries.

Sample Dataset

Customer_ID	Age	Gender	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

Tasks

A. Handling Missing Values

- Identify missing values in **Age**, **Monthly_Spend**, and **Purchase_Frequency**
- Apply at least **two imputation techniques** (e.g., Median Imputation, k-NN Imputation)
- Compute the imputed values and justify the selected method

B. Outlier Detection and Treatment

- Detect outliers in **Age** and **Monthly_Spend** using the **Z-score method**
- Identify records containing extreme values (e.g., Age = 105, Spend = 500)
- Apply suitable treatment methods (**capping** / **transformation** / **removal**) and justify

C. Categorical Data Standardization

- Identify inconsistencies in **Gender** (M, Male, male, FEMALE, etc.)
- Convert all entries into a standardized format (**Male** / **Female**)

Q2. Covariance & Data Reduction in E-commerce Analytics (L4, L5)

An **e-commerce platform** is building a predictive model to classify high-value customers. The dataset includes several correlated financial and behavioral attributes, leading to redundancy and increased computational cost.

Sample Dataset

Cust_ID	Age	Income (₹K)	Purchase_Value (₹K)	Credit_Score	Orders	Avg_Balance (₹K)	EMI (₹)	Customer_Class
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

Tasks

A. Covariance Analysis

- Compute covariance between **Income and Purchase_Value** (show steps or partial calculation)
- Interpret whether the relationship is **positive or negative**
- Explain how covariance helps in identifying feature relationships

B. Feature Selection

- Identify redundant attributes using **correlation or domain knowledge**
- Select an optimal subset of features for modeling
- Justify your selection

C. Data Transformation

- Standardize the dataset (excluding **Cust_ID** and **Customer_Class**)
- Explain why normalization/standardization is required

Q3. Data Transformation & Discretization in Education Analytics (L3, L4, L5)

An **education analytics team** is developing a model to predict student performance. Continuous variables such as study hours and test scores need to be discretized to improve interpretability and decision-making.

Sample Dataset

Student_ID	Study_Hours	Test_Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

Tasks

A. Equal-Width Binning

- Apply **3 bins** for Study_Hours and Test_Score
 - Calculate bin width and define intervals
-

B. Equal-Frequency Binning

- Divide data into **3 bins with equal records**
 - Assign values accordingly
-

C. Concept Hierarchy Construction

- Study_Hours → Low / Medium / High
- Test_Score → Poor / Average / Excellent

4. Use the two methods below to normalize the following group of data: 200, 300, 400, 600, 1000
- min-max normalization by setting min = 0 and max = 1
 - z-score normalization

5. Data Transformation in Online Shopping Behavior Analysis (L3, L4, L5)

An **e-commerce company** is analyzing customer engagement by measuring the **time spent (in seconds)** on product pages. The duration varies significantly—from a few seconds to more than an hour—resulting in highly skewed data.

To improve the effectiveness of **customer segmentation using clustering techniques**, the dataset must be properly transformed and normalized.



Sample Dataset: Visit Duration

Customer_ID Visit_Duration (seconds)

C1	15
C2	30
C3	45
C4	60
C5	120
C6	300
C7	600
C8	900
C9	1800
C10	3600



Tasks

A. Equal-Frequency Binning

- Divide the dataset into **3 bins** with approximately equal number of records
 - Assign each value to its respective bin
 - Analyze how effectively this method groups short and long visit durations
-

B. Z-Score Normalization

- Compute the **mean and standard deviation** of visit duration
- Apply Z-score normalization:

$$Z = \frac{X - \mu}{\sigma}$$

- Discuss how this method handles **extreme values** (e.g., 1800, 3600 seconds)
-

C. Min-Max Normalization

- Apply Min-Max normalization using range **[0,1]**:

$$X = \frac{X - X_{min}}{X_{max} - X_{min}}$$

- Transform all values accordingly
- Compare results with Z-score normalization in terms of **scaling and sensitivity to outliers**

ANSWER:

1Q. Data Preprocessing in Retail Analytics (L3, L4)

A. Handling Missing Values

i. Missing Values Identified

Column Missing Record

Age C2

Monthly_Spend C3

Purchase_Frequency C5

ii. Two Imputation Techniques

Median Imputation

Age = 36

Monthly_Spend = 36.5

Purchase_Frequency = 10

k-NN Imputation

Age $\approx$ 27

Monthly_Spend $\approx$ 40

Purchase_Frequency $\approx$ 9

iii. Justification

Median Imputation: Best for numerical data with outliers because it is not affected by extreme values.

k-NN Imputation: Uses similar records to estimate missing values, giving more accurate results.

B. Outlier Detection and Treatment

i. Z-score Method

Outliers detected:

Age = 105 (C4)

Monthly_Spend = ₹500K (C4)

ii. Records with Extreme Values

Customer_ID Outlier

C4 Age = 105, Spend = ₹500K

iii. Treatment

Age (105): Cap to the maximum acceptable age (e.g., 80) or remove if it is a data entry error.

Monthly_Spend (500): Cap to a reasonable upper limit (e.g., 300) or apply log transformation.

Justification:

Capping reduces the impact of extreme values while retaining the record for analysis.

C. Categorical Data Standardization

i. Inconsistencies

M

Male

male

F

Female

FEMALE

ii. Standardized Format

Original Standardized

M Male

Male Male

male Male

F Female

Female Female

FEMALE Female

This ensures consistency and improves data quality for analysis and machine learning.

2Q- Covariance & Data Reduction in E-commerce Analytics (L4, L5)

A. Covariance Analysis

i. Covariance between Income and Purchase_Value

Income (₹K): 40, 60, 85, 120, 45, 75, 150, 70

Purchase_Value (₹K): 100, 150, 210, 320, 110, 180, 360, 160

Mean Income = 80.63

Mean Purchase_Value = 198.75

Using the covariance formula:

$$\text{Cov}(X, Y) = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{n-1}$$

Covariance ≈ 3275.9

ii. Interpretation

Covariance is positive (+3275.9).

As Income increases, Purchase_Value also increases.

iii. Importance of Covariance

Identifies relationships between features.

Helps detect highly related variables.

Useful for feature selection and dimensionality reduction.

B. Feature Selection

i. Redundant Attributes

Income $\leftrightarrow$ Purchase_Value

Income $\leftrightarrow$ Avg_Balance

Purchase_Value $\leftrightarrow$ EMI

These attributes are highly correlated.

ii. Selected Features

Age

Income

Credit_Score

Orders

Customer_Class (Target)

iii. Justification

Removes redundant features.

Reduces computation time.

Improves model efficiency and avoids overfitting.

C. Data Transformation

i. Standardization

Apply Z-core Standardization (excluding Cust_ID and Customer_Class).

Formula

$$Z = \frac{X - \mu}{\sigma}$$

Standardize:

Age
Income
Purchase_Value
Credit_Score
Orders
Avg_Balance
EMI

ii. Why Standardization is Required

Brings all features to the same scale.
Prevents large-value features from dominating.
Improves the performance of machine learning algorithms.
Speeds up model training and increases accuracy.

3Q- Data Transformation & Discretization in Education Analytics (L3, L4, L5)**A. Equal-Width Binning****i. Apply 3 Bins**

Study_Hours
Minimum = 2
Maximum = 11
Bin Width = $(11 - 2) / 3 = 3$

Bins:

Bin 1: 2 – 5
Bin 2: 5 – 8
Bin 3: 8 – 11

Test_Score

Minimum = 40
Maximum = 100
Bin Width = $(100 - 40) / 3 = 20$

Bins:

Bin 1: 40 – 60
Bin 2: 60 – 80
Bin 3: 80 – 100

B. Equal-Frequency Binning

Study_Hours
Bin Values

Bin 1 2, 3, 4, 5
Bin 2 6, 7, 8
Bin 3 9, 10, 11

Test_Score

Bin Values

Bin 1 40, 50, 55, 65
Bin 2 75, 85, 90
Bin 3 95, 98, 100

C. Concept Hierarchy

Study_Hours

Low: 2–4
Medium: 5–7
High: 8–11

Test_Score

Poor: 40–59
Average: 60–79
Excellent: 80–100

4. Normalization

Given Data:

200, 300, 400, 600, 1000
(a) Min-Max Normalization (0 to 1)

Formula:

$$X' = \frac{X - \text{Min}}{\text{Max} - \text{Min}}$$

Min = 200, Max = 1000
Value Normalized
200 0.00
300 0.125
400 0.25
600 0.50
1000 1.00

(b) Z-Score Normalization

Formula:

$$Z = \frac{X - \mu}{\sigma}$$

Mean (μ) = 500
Standard Deviation (σ) $\approx$ 282.84

Value Z-Score

200 -1.06
300 -0.71
400 -0.35
600 0.35
1000 1.77

5Q- Data Transformation in Online Shopping Behavior Analysis (L3, L4, L5)

A. Equal-Frequency Binning

i. Divide into 3 Bins

10 records → Approximately 3, 3, and 4 records

Bin Visit Duration (seconds)

Bin 1 15, 30, 45

Bin 2 60, 120, 300

Bin 3 600, 900, 1800, 3600

ii. Assign Values

Bin 1: C1, C2, C3

Bin 2: C4, C5, C6

Bin 3: C7, C8, C9, C10

iii. Analysis

Groups similar visit durations together.

Separates short, medium, and long visits.

Useful for clustering customers.

B. Z-Score Normalization

i. Mean and Standard Deviation

Mean (μ) = 747 seconds

Standard Deviation (σ) $\approx$ 1113.95 seconds

ii. Z-Score Formula

$$Z = \frac{X - \mu}{\sigma}$$

Visit Duration	Z-Score
15	-0.66
30	-0.64
45	-0.63
60	-0.62
120	-0.56
300	-0.40
600	-0.13
900	-0.14
1800	-0.94

3600

-2.56

iii. Discussion

Centers data around 0.

Large values (1800, 3600) have high positive Z-scores.

Helps identify extreme values (outliers).

C. Min-Max Normalization

i. Formula

$$X' = \frac{X - \text{Min}}{\text{Max} - \text{Min}}$$

Min = 15, Max = 3600

ii. Normalized Values

Visit Duration	Min-Max Value
15	0.000
30	0.004
45	0.008
60	0.013
120	0.029
300	0.080
600	0.163
900	0.247
1800	0.498
3600	1.000

iii. Comparison

Min-Max Normalization: Scales all values between 0 and 1, but is more sensitive to outliers.

Z-Score Normalization: Centers data around 0 with standard deviation 1, making it better for handling extreme values like 1800 and 3600 seconds.